

NIC KANE

Entry-Level Data Scientist | M.S. Statistical Data Science Candidate

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PROFESSIONAL SUMMARY

Entry-level Data Scientist and M.S. Statistical Data Science candidate with experience in statistical modeling, machine learning, classification, and time series forecasting using Python and R. Nearly nine years of teaching experience strengthens technical communication and the ability to translate complex analytical insights for diverse audiences

TECHNICAL SKILLS

Languages & Tools: Python (pandas, NumPy, PyTorch, scikit-learn), R, SQL, Excel, Tableau, Power BI, C, LaTeX, R Markdown

Methods: Regression, Classification, LDA/QDA, KNN, Random Forests, Boosting, Ridge/Lasso, Time Series (ARIMA, SARIMA, ETS), Cross-Validation, ROC/AUC, AIC Model Selection

APPLIED DATA SCIENCE PROJECTS

Predicting IMDb Movie Ratings from Metadata

- Built and compared linear regression, ridge/lasso, support vector regression, and random forest models in Python (pandas, scikit-learn) to predict IMDb average rating from runtime, release year, and genre, using a merged dataset of 100K+ films.
- Engineered features through extensive data cleaning: missingness analysis, one-hot encoding of multi-label genres, log-transformation of skewed vote counts, and outlier filtering on runtime.
- Tuned models via GridSearchCV and cross-validation; identified genre (Documentary, Horror) and runtime as the most consistent predictors across all model types.
- Achieved best performance with a tuned random forest (RMSE = 1.20), improving over baseline OLS while preserving interpretable variable importance rankings.

Time Series Analysis of U.S. Vinyl Record Sales

- Analyzed longitudinal sales data to identify trends and assess stationarity using ACF/PACF.
- Built and compared ARIMA and exponential smoothing models using AIC and residual analysis.
- Generated short- and long-term forecasts with confidence intervals and interpreted forecast uncertainty — applicable to demand forecasting and business analytics.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, Prelude to Calculus, Calc I Lab | *San Francisco State University*

Aug 2025 – Dec 2026

- Taught a preparatory mathematics course for STEM majors building readiness for calculus-level coursework using small-group collaborative learning.
- Designed and administered a midterm exam, final exam, and cumulative project to assess student readiness.
- Translated complex quantitative concepts into clear, structured explanations — a skill directly transferable to communicating model results and analytical findings.

Instructor, Mathematics & Music | *Fusion Academy, San Francisco Bay Area*

Feb 2019 – Aug 2026

- Delivered individualized mathematics instruction from 6th grade through calculus, adapting explanations to each student's learning style.
- Communicated complex quantitative concepts clearly to students with diverse academic backgrounds and learning needs.
- Maintained detailed progress documentation and collaborated with students, parents, and administrators.

Additional Teaching & Leadership Experience: Math & Physics Instructor, Lydian Academy (2018–2019); Senior Leader & Math Workshop Facilitator, Cal Poly San Luis Obispo (2015–2018) — supported quantitative instruction, workshop facilitation, and facilitator coordination.

EDUCATION

M.S. in Statistical Data Science

Expected December 2026

San Francisco State University, San Francisco, CA

B.S. in Electrical Engineering, Minor in Mathematics

2018

Cal Poly, San Luis Obispo, CA